Diagnosing Mitigation Need: Controlled Shortages and Signal-Matched Interventions in Histopathology Patch Classification

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Abstract

A class can be underserved for several distinct reasons, and the count that marks it as rare does not say which. In computational pathology, raw sample counts hide at least one further shortage — few labelled patients or slides behind many informative patches — and say nothing about how separable a class is under the available representation or how much variation it spans. This benchmark uses training allocation as a controlled way to induce such a shortage, and then asks which signal a mitigation method should read to repair it: how would the same prediction task change if one fixed training budget were distributed evenly across classes rather than concentrated in head classes, which signal profile does the resulting condition actually present, and does a method that reads the signal matching that profile recover more of the damage than one that reads a different signal? CAMELYON16, TCGA-UT, BRACS, and PANDA are studied in patch classification using frozen Virchow2 features. Patient- or case-disjoint partitions, natural validation and test cohorts, model family, and each method's optimizer-update policy remain fixed across conditions. Five size-matched controlled training sets vary either the allocation of patches among classes or the number of patients supplying an identical allocation: balanced, moderate, and severe sets over a concentrated patient pool, plus balanced and severe sets over the full eligible patient pool. Balanced accuracy and tail-group macro negative log likelihood measure discrimination and probability quality. Mitigation recovery is interpreted only after a prespecified gate establishes a meaningful allocation-induced deficit. A five-member family of methods — one per signal, all a mean-one class weight on unmodified cross-entropy — makes the signal contrast interpretable at a fixed intervention level. Exploratory analyses then relate damage and recovery to achieved allocation severity, independent-evidence shortage, support–difficulty alignment, and representational-diversity shortage. The benchmark is scoped to patch classification, where each prediction unit carries its own label.

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1 Introduction

Choosing a mitigation method requires first identifying what limits a class. In histopathology patch classification, an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity. These shortages are not interchangeable. Thousands of patches from a few patients may provide less independent evidence than a smaller patch set drawn from many patients, while neither count reveals how difficult a class is to separate or how much variation it contains. A method that responds to one shortage may therefore be ineffective when another shortage limits performance.

This benchmark asks when such a shortage causes enough damage to warrant mitigation and whether recovery improves when a method reads the signal that identifies that shortage. Training allocation provides the controlled intervention needed to answer these questions. For the same task and total training budget, examples are distributed either evenly across classes or concentrated in head classes. A size-matched balanced cross-entropy (CE) model provides the reference. Validation and test cohorts retain their natural class distributions, so the resulting contrast reflects the training allocation rather than a change in the evaluation population.

Allocation is thus a way to create controlled variation in evidence, not the benchmark's main explanatory focus. Each condition is instead characterized by its realized *signal profile*: class prevalence, nominal patch support, independent patient or slide support, class difficulty, and representational diversity. This profile is measured rather than inferred from the nominal imbalance ratio. The companion methods report defines the signal taxonomy and locates each tested method within it (Queisler, 2026b); the present benchmark tests whether matching method to signal matters empirically.

The study covers CAMELYON16, TCGA-UT, BRACS, and PANDA using frozen Virchow2 patch features. Patient- or case-disjoint partitions, model family, method-specific update policy, and natural validation and test cohorts remain fixed while the allocation of a common training budget or its patient breadth changes. Concentrated balanced, moderate, and severe allocations are paired with balanced and severe allocations over the full eligible patient pool. To separate signal choice from intervention mechanism, the confirmatory family contains five methods that all apply a mean-one class weight to unmodified CE. Their weights differ only in whether they use prevalence, nominal support, independent support, difficulty, or diversity.

This design yields three research questions:

RQ1. Which shortage does controlled allocation create, and what does it damage? Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

RQ1 first establishes whether a recoverable deficit exists and which signals characterize it. RQ2 is evaluated only after this deficit passes a prespecified gate. RQ3 then examines why deficit size and recovery headroom vary; it is an exploratory association analysis, not another method ranking.

The benchmark is limited to patch classification because directly labelled patches allow nominal patch support to change within a fixed pool of patients or slides. At whole-slide level, reallocating labelled slides changes nominal and independent support together, making the two signals harder to separate. Dataset provenance and detailed method objectives remain in the companion reports (Queisler, 2026a,b). The next section defines the prediction and evidence units on which the experimental design rests.

2 Study Setting

This benchmark covers one prediction regime: patch classification predicts a label for one directly labelled image crop, so every training example carries its own target. The terms *patch* and *tile* both denote a 256×256 crop, but *patch* is used throughout.

Table 1 gives the patch target and patient/case grouping used by each dataset.

Dataset	Patch-level example and target	Patient/case grouping
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI's cancer type	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	Biopsy (one WSI)

Table 1: Dataset-specific definitions of patch-level examples. TCGA-UT's eligible target set omits two of the 32 annotated cancer types; Appendix A states which classes and why.

With the prediction examples, targets, and patient/case groupings established, the next section follows the controlled design from eligible examples to controlled model comparisons.

3 Controlled Experimental Design

Figure 1 gives the complete design before its steps are considered separately. Eligible examples are divided into training, validation, and test partitions by partitioning unit; training support is then formed into a natural anchor and five controlled training sets; controlled model comparisons are finally evaluated on unchanged natural validation and test cohorts. Within a comparison, either class allocation changes at fixed patient-pool breadth or patient-pool breadth changes at fixed class allocation.

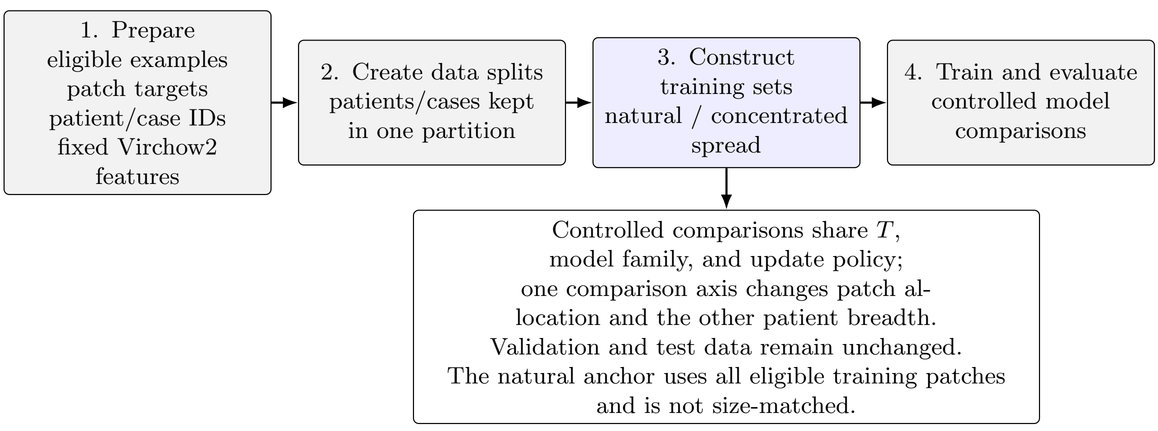

Figure 1: Controlled benchmark design. Gray boxes define shared evidence, partitioning, and evaluation; the blue box contains the training-support intervention.

3.1 Eligible examples and data partitions

A patch becomes an *eligible example* only if it has a valid target and patient/case identifier, satisfies its dataset's tissue and annotation rules, and has readable fixed features. Eligibility is determined once, before partitions or training conditions are constructed. Every eligible patch is embedded once with a frozen Virchow2 encoder; concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). Appendix A gives the complete exclusion, preprocessing, and provenance rules.

Within each data split, a *partition* is one of three mutually exclusive subsets: training, validation, or test. A *partitioning unit* is the patient or participant where available and the biopsy otherwise. Exactly three partitioning-unit-disjoint data splits are derived from the eligible examples, and every patch from one unit follows the same assignment. Imbalance manipulation is confined to the training partition, so every condition retains the same natural validation and test partitions. No held-out example is deleted, duplicated, or reweighted according to training severity. Validation selects hyperparameters and checkpoints; test evaluation uses every eligible patch at one fixed prevalence.

3.2 Training-support construction

Six training sets separate the descriptive anchor from the controlled experiment. The *natural anchor* uses all eligible training patches and describes performance at the dataset's observed prevalence. Five controlled sets share the same total number of patches T. Over the concentrated patient pool, balanced, moderate, and severe allocations target $\rho = 1$, $\rho = 10$, and $\rho = 100$. The paired spread sets place the balanced and severe patch allocations over the full eligible patient pool. Appendix A.2 specifies how the concentrated and spread pools are formed.

These five controlled sets define four deprived conditions:

1. concentrated moderate compared with concentrated balanced;
2. concentrated severe compared with concentrated balanced;
3. concentrated balanced compared with balanced spread; and
4. severe spread compared with balanced spread.

The first, second, and fourth comparisons change nominal class allocation while holding patient-pool breadth fixed. The third changes patient breadth while preserving each class's patch count. The concentrated balanced set therefore serves as the reference for nominal deprivation and as the deprived set for independent-support deprivation. All five controlled sets carry the full method roster; the natural anchor carries CE only.

Figure 2 makes the nominal intervention concrete using one BRACS split. The natural allocation shows the source distribution but is not size-matched. Spread variants are not drawn separately because their per-class patch counts are identical to the corresponding balanced and severe bars; only the number of contributing patients differs.

The example shows why the natural anchor and controlled conditions must not be conflated. Natural BRACS is already imbalanced ($\rho_{\text{achieved}} = 5.90$) and contains 217,399 training patches, almost 13 times the controlled budget. The controlled allocations instead hold $T = 16{,}796$ fixed; their achieved ratios are 1.00, 10.01, and 100.47. Other tail assignments map the same moderate and severe support profiles to different semantic classes, while spread variants change patient breadth without changing T or the severity profile.

The shared T is selected jointly with severity rather than from size alone. Construction retains the largest feasible total for which every tail assignment realizes moderate $\rho_{\text{achieved}} \in [9, 11]$ and severe $\rho_{\text{achieved}} \in [90, 110]$. Appendix A specifies the search and feasibility rules.

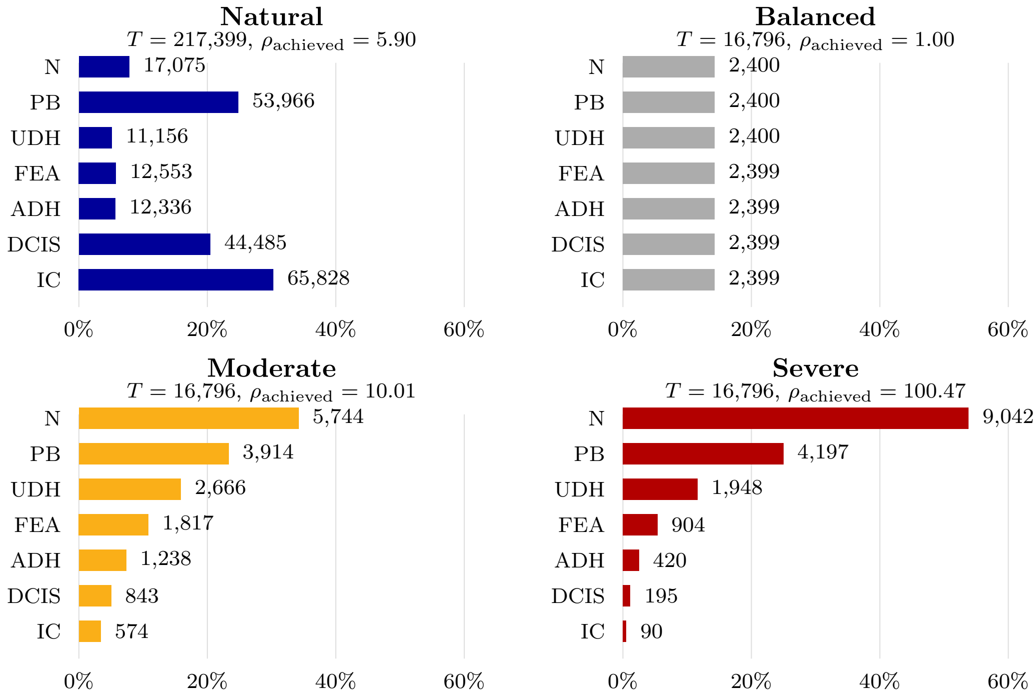

Figure 2: Realized BRACS patch allocations. Bars show each class's share of training patches; labels give realized patch counts. Moderate and severe use the native tail assignment. The natural anchor uses all eligible training patches, while all five controlled sets share $T = 16{,}796$. Balanced spread and severe spread have the same per-class patch counts as the corresponding bars and differ only in contributing-patient breadth. Class abbreviations are N (normal), PB (pathological benign), UDH (usual ductal hyperplasia), FEA (flat epithelial atypia), ADH (atypical ductal hyperplasia), DCIS (ductal carcinoma in situ), and IC (invasive carcinoma).

Within each controlled allocation, severity determines how the fixed total T is distributed across support ranks. Binary tasks have one head and one tail. Multiclass tasks place intermediate supports at equal intervals on a logarithmic scale, as the BRACS profiles in Figure 2 illustrate. A moderate or severe construction that collapses to the balanced allocation is marked degenerate and is not pooled with genuine imbalanced conditions.

The target ratio $\rho_{\text{target}} \in \{1, 10, 100\}$ specifies the intended head–tail contrast. Because a training set contains whole patches and each class has a limited eligible inventory, that ratio may not be realized exactly. The achieved ratio is therefore recomputed from every training set. For realized class supports $N_1, \ldots, N_K$, it is

$$\rho_{\text{achieved}} = \frac{N_{\max}}{N_{\min}}. \tag{1}$$

This ratio records the most extreme head–tail contrast. The complementary normalized-entropy deficit

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^{K} p_c \log p_c}{\log K}, \qquad p_c = \frac{N_c}{\sum_j N_j}. \tag{2}$$

uses the complete class distribution and is normalized for class count. It equals zero for a balanced allocation and increases as support becomes concentrated. Both summaries are computed from realized rather than requested counts. Table 2 separates overall training support from condition-specific imbalance. Its upper part contrasts the natural anchor with the controlled realizations; its lower part shows how their class distributions differ.

Dataset	Natural			Controlled		
	Units	Slides	Patches	Units	Slides	Patches
BRACS	103	262–273	193,010–217,704	78–89	173–233	13,982–27,202
CAMELYON16	264	264	2,139,913–2,234,906	103–118	103–118	28,810–36,127
PANDA	7,358	7,358	3,141,372–3,161,828	6,025–6,080	6,025–6,080	1,099,847–1,120,125
TCGA-UT	4,983	6,055–6,100	1,116,020–1,121,580	671–917	699–933	36,988–52,204

Dataset	Metric	Natural	Balanced	Moderate	Severe
BRACS	ρ_{achieved}	5.722–8.297	1.000–1.001	10.003–10.007	100.301–100.467
	$1 - H_{\text{norm}}$	0.118–0.143	0.000	0.128	0.353
CAMELYON16	ρ_{achieved}	22.453–33.608	1.000	9.998–10.001	99.913–100.088
	$1 - H_{\text{norm}}$	0.746–0.811	0.000	0.560–0.561	0.920
PANDA	ρ_{achieved}	4.645–4.735	1.000	10.000	99.996–100.003
	$1 - H_{\text{norm}}$	0.326–0.332	0.000	0.561	0.920
TCGA-UT	ρ_{achieved}	17.850–18.395	1.001	9.992–10.003	90.107–100.500
	$1 - H_{\text{norm}}$	0.075–0.076	0.000	0.061	0.174–0.179

Table 2: Overall realized training support (top) and imbalance (bottom). Units are partitioning units. Entries give the minimum and maximum across three frozen splits; moderate and severe imbalance ranges additionally span all retained tail assignments. The controlled support columns summarize balanced, moderate, and severe realizations together: their patch total is shared within a split, while the selected units and slides can vary with allocation. Counts are unique within each realization, even when one unit contributes more than one class. A single value indicates an invariant quantity.

3.3 Tail assignments

A *tail assignment* determines which semantic class receives each support rank. The native assignment follows the dataset's clinical or ordinal class order. The difficulty-aligned assignment gives less support to harder classes, compounding scarcity and difficulty; the reversed assignment gives harder classes more support. Duplicated mappings are omitted transparently. Figure 3 illustrates how the same severity profile yields different semantic tails.

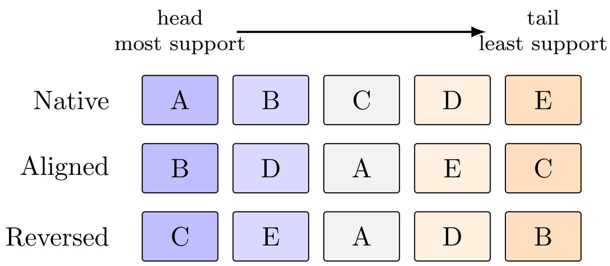

Figure 3: Illustrative tail assignments for a five-class target. Columns are support ranks fixed by the severity profile. Difficulty increases from B to C: aligned compounds difficulty with scarce support, while reversed compensates it.

Detailed support floors, allocation equations, infeasibility rules, seed families, and locked controls are specified in Appendix A and Table 5. The next section asks how controlled model comparisons turn that evidence into damage and recovery estimates.

4 Measuring Damage and Recovery

Section 3 established the controlled training conditions and the evidence held fixed across them. This section follows each controlled comparison from fitted models to an inferential claim. It first identifies the compared methods and their shared execution controls, then defines the endpoints used to detect imbalance-induced damage, measures mitigation recovery only where such damage is demonstrated, and finally quantifies uncertainty and tests the prespecified signal contrast. RQ1 and RQ2 are therefore answered in a deliberately asymmetric sequence: balanced CE is compared with imbalanced CE first, and only when that contrast shows a meaningful deficit on discrimination or calibration is a mitigation method interpreted in terms of recovery on the affected endpoint.

For intuition, suppose balanced CE attains balanced accuracy 0.72, imbalanced CE attains 0.62, and a mitigation method attains 0.68. Imbalance caused a 0.10 deficit and the method regained 0.06, or 60% of the demonstrated damage. This normalization distinguishes repair of an imbalance effect from a generic method improvement. Exact estimands and gates appear after the endpoints they use.

4.1 Compared methods and shared execution controls

Table 3 lists only the methods evaluated in this benchmark, grouped by the mitigation signal each one reads. CE is the size-matched no-mitigation reference. The roster is fitted in all five controlled training sets, which supply the paired contrasts required for mitigation claims; the natural anchor carries the CE reference only. Method objectives, tested adaptations, ablations, defaults, and fidelity qualifications are defined exclusively in the companion methods report (Queisler, 2026b).

Signal	Tested methods	Intervention
None (reference)	CE	None
Prevalence	Balanced sampling; weighted CE; train-time and post-hoc logit adjustment; cRT	Sampling; loss weight; logits and prior correction; classifier retraining
Nominal support	Class-balanced CE	Loss weight
Independent support	Independent-support weighted CE	Loss weight
Difficulty	Pilot-difficulty weighted CE	Loss weight
Diversity	Semantic-scale weighted CE	Loss weight (dynamic)
Nominal support and difficulty	Focal loss; CFAL	Loss weight; prototype head and margin loss
Prevalence (hybrid)	CE + soft F1; CE + soft MCC; OKO set learning	Sampler and auxiliary loss; set construction and set loss

Table 3: Prespecified patch-regime method roster, grouped by mitigation signal following the companion methods report, with intervention level retained as a second column so the two axes stay visibly orthogonal.

All controlled fits retain the frozen encoder, model family, optimizer family, batch definition, augmentation, and numerical precision. They share an example-presentation budget; method- and configuration-specific update counts are derived from it (Appendix D). cRT adds a classifier-only second stage and post-hoc logit adjustment reuses CE. Hyperparameters and checkpoints use natural validation only and are locked before confirmation and test evaluation.

Figure 4 places these steps in execution order. It also separates the validation search that is self-contained from the phase that cannot start until a condition's CE search has resolved, which is what fixes the order in which runs may be launched.

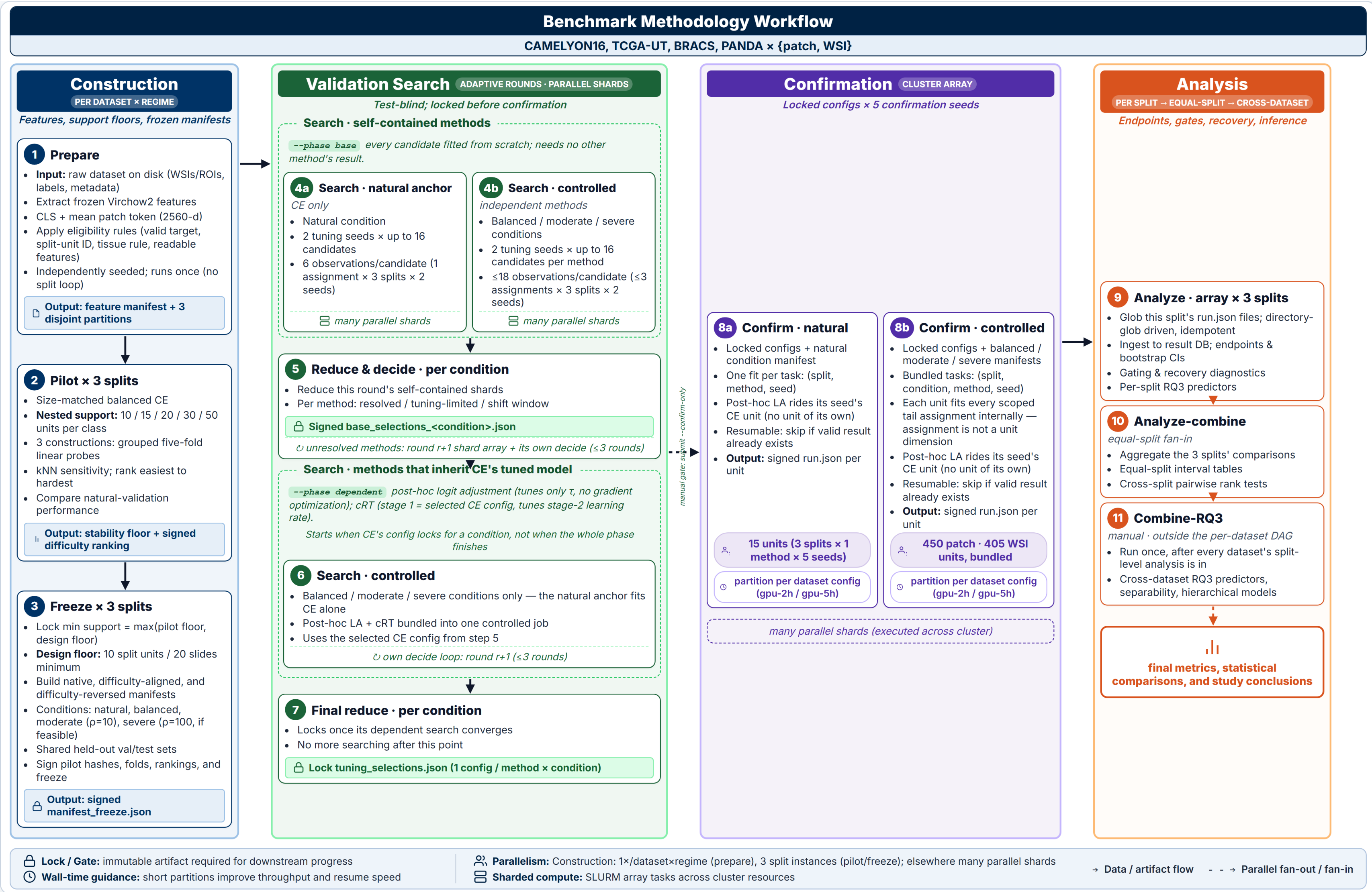

Figure 4: Operational workflow of the controlled class-imbalance benchmark.

This separation matches fitted methods by data exposure rather than update count. With models fixed, evaluation can ask which endpoint was damaged and how much recovered.

4.2 Evaluation endpoints

Damage and recovery are evaluated on two complementary axes: discrimination measures whether the predicted class is correct, whereas probability quality measures whether the complete predicted distribution is reliable. Both axes preserve the independent patient or case as the unit of evidence.

Discrimination and aggregation. The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Overall accuracy is descriptive. The primary, gate-defining estimator is case-macro: the metric contribution is first aggregated within each partitioning unit, so a unit's contribution to a class counts once regardless of how many patches or slides it supplies, and heavily tiled cases do not dominate. Patch-micro estimates, which instead treat eligible patches directly as predictions while uncertainty remains clustered by partitioning unit, are reported as a secondary view.

Probability quality. Probability-quality metrics ask whether the complete predicted distribution is useful, not only whether its largest entry names the correct class. Tail-group macro negative log likelihood (NLL) is the co-primary calibration endpoint because it gives each tail class equal weight and penalizes confident errors. Because NLL uses the natural logarithm, its values and differences are measured in nats. A nat is the unit of information obtained with logarithms of base e, just as a bit is obtained with logarithms of base 2. Natural-prevalence NLL, macro NLL, multiclass Brier score, expected calibration error (ECE), and reliability diagrams provide secondary views. ECE uses 10 fixed equal-width bins and remains secondary because small classwise samples make bin estimates unstable.

Discrimination uses balanced-decision logits. The CE calibration gate uses target-prior-corrected probabilities before temperature scaling; such a correction is defined only for CE and the two logit-adjustment variants, which is sufficient because the gate is computed from CE runs alone. Method-specific probability-quality endpoints instead follow the per-method score semantics fixed in Appendix B: methods without an explicit prior correction are scored on normalized raw scores, so their tail-NLL changes are probability-quality changes under those semantics and not, in general, evidence of recovered calibration. Temperature-scaled outputs are secondary and never enter a recovery gate. Appendix B defines every probability-quality equation, clipping at 10^{-12}, score variants, and fixed-bin ECE.

4.3 From imbalance deficit to mitigation recovery

Recovery is meaningful only when imbalance has first caused a measurable loss. The analysis therefore proceeds in two stages: compare size-matched balanced and imbalanced CE, then evaluate mitigation only on an endpoint that shows sufficiently large and statistically supported harm. One *comparison unit* is a fixed dataset, imbalance severity, and tail assignment. A *gate* is the prespecified eligibility rule that routes such a unit to the relevant recovery analysis; it is not a learned model component.

For a higher-is-better metric M, define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$D_M = M(\text{balanced CE}) - M(\text{imbalanced CE}),$$
$$R_M = \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \tag{3}$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T, methods that only reweight or recalibrate the imbalanced training set cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M: it has no size-matched balanced counterpart, so neither a deficit nor a recovery fraction is defined there. Fitting mitigation methods on it would therefore yield differences confounded by training volume as well as allocation, and the anchor is restricted to its CE reference for that reason.

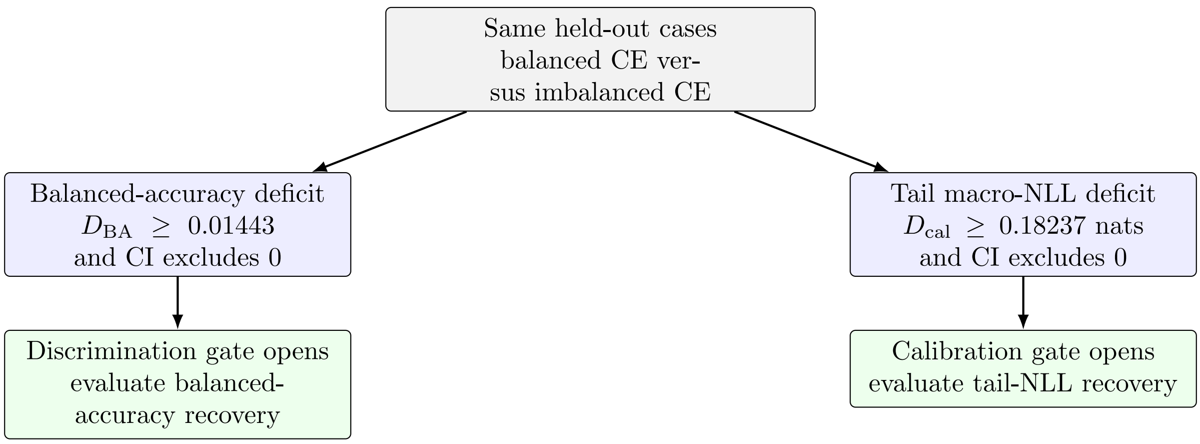

Figure 5: Routing from an imbalance deficit to mitigation analysis. A comparison unit can enter through either gate or both. If neither rule is met, its CE deficits are still reported, but a recovery ratio is not interpreted. CI denotes the paired 95% confidence interval.

Figure 5 shows the two routes. The *discrimination gate* opens when the CE balanced-accuracy deficit is at least 0.01443 and its paired 95% confidence interval excludes zero. The *calibration gate* opens when the CE tail-group macro-NLL deficit, $\mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$, is at least 0.18237 nats with a paired 95% confidence interval excluding zero. Both rules are computed from CE runs on the same held-out cases before any mitigation method is compared.

Both thresholds are derived, not assumed. The discrimination threshold is $\max(0.01, 2\sigma_{\mathrm{seed}})$, where the floor 0.01 is the support pilot's own balanced-accuracy materiality constant (Appendix A.2) and σ_{seed} is the largest across datasets and splits of the confirmation-seed standard deviation of balanced-condition CE case-macro balanced accuracy; the noise-floor term binds ($\sigma_{\mathrm{seed}} = 0.00721$, BRACS). The calibration threshold is 0.05 nats, a 5% relative reduction in the geometric-mean probability assigned to true tail-class labels, unless the analogous tail-NLL σ_{seed} exceeds 0.025, in which case it is raised to $2\sigma_{\mathrm{seed}}$; here it does ($\sigma_{\mathrm{seed}} = 0.09119$, CAMELYON16), so the calibration threshold is raised accordingly. A comparison unit entering through the discrimination gate is evaluated for balanced-accuracy recovery; one entering through the calibration gate is evaluated for tail-NLL recovery under the method-specific score semantics of Table 8; it may enter through both.

The two routes preserve an important distinction. Strong frozen features may leave balanced accuracy nearly intact while imbalance still biases tail probabilities, in which case a single discrimination rule would exclude methods designed to correct class priors. Conversely, the gates avoid unstable recovery ratios when no meaningful damage exists to recover. All CE deficits on both axes remain reported, including units that fail both rules. For the lower-is-better calibration axis, the sign-corrected deficit $D_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$ is the denominator and the method's tail-NLL improvement over imbalanced CE is the numerator, so Equation (3) retains the same interpretation. Recovery for other secondary metrics is descriptive and does not reopen a comparison unit that fails both gates.

4.4 Addressing uncertainty

The deficit and recovery estimates in Section 4.3 summarize a finite held-out cohort rather than fixed population quantities. Their uncertainty has two sources: different patients or cases could produce different estimates, and repeated model fitting introduces algorithmic variation. Multiple patches from the same patient or case do not provide equivalent independent evidence, just as repeated fits do not create additional patient evidence.

Uncertainty is therefore estimated by resampling patients or cases as complete units and repeating the relevant comparisons. The same resampled cohort and corresponding model-initialization draws are used for the balanced CE, imbalanced CE, and mitigation results within each controlled comparison. For intuition, suppose one resample gives a particular patient greater influence. That patient's predictions receive the same greater influence when balanced CE, imbalanced CE, and the mitigation method are evaluated. The deficit and recovery fraction therefore remain comparisons on the same reweighted cohort rather than comparisons between differently sampled groups.

Repeating this process and recomputing the complete deficit and recovery fraction shows how much both estimates vary with cohort composition and model initialization, producing their 95% confidence intervals. Data splits are averaged as design repetitions rather than treated as independent cohorts. Appendix B specifies the resampling construction, validity checks, support diagnostics, and interval calculation.

4.5 Interpreting the evidence

The preceding subsections establish how models are compared, which endpoints record damage, when recovery is interpretable, and how uncertainty is represented. RQ1 is answered by the signed CE deficit and its confidence interval: together they show the direction and magnitude of the allocation effect, while the gate identifies damage large and certain enough to support a recovery analysis. RQ2 begins only for those gate-passing comparisons. A method's improvement over imbalanced CE and its recovery fraction then show how much of the demonstrated damage was repaired. Raw effects and confidence intervals remain the main evidence; recovery fractions are normalized summaries rather than test statistics.

The method roster creates many related method-by-condition comparisons. Treating every result as an independent primary test would increase the chance of a false-positive claim, so the benchmark prespecifies a limited confirmatory family whose tests are adjusted jointly. The confirmatory part of RQ2 asks whether signal choice matters when the intervention mechanism is held fixed. It therefore compares the five weighted-CE methods in Table 3, whose weights use prevalence, nominal support, independent support, difficulty, or diversity while leaving the CE mechanism unchanged. Differences among them can consequently be interpreted as evidence about the signal rather than the intervention. Matching is made explicit: a prespecified rule labels each comparison unit with the dominant shortage in its realized signal profile, and one direct contrast tests the method reading that signal against the unmatched members of the family, so the matching claim rests on a between-method comparison rather than on which methods separately beat imbalanced CE. Units whose two leading shortages are too close to separate are labelled ambiguous and contribute descriptively only. Within each dataset, paired tests support the gate-passing comparisons and Holm adjustment controls the joint family-wise error of the method-versus-CE tests and the matched contrasts together. Other roster members retain the same effect estimates and confidence intervals but remain exploratory. Appendix B.3 specifies the tests, multiplicity family, and protocol revision; Appendix D.2 gives method-specific diagnostic and tuning details.

Section 4 therefore produces three outputs from each controlled comparison: whether damage occurred, how large it was, and how much of it was recovered when recovery was defined. RQ1 and RQ2 estimate these quantities within a dataset and task. The remaining question is why damage magnitude and recovery vary across datasets, severities, and tail assignments. Section 5 models those two continuous outputs and retains damage occurrence only as the gate that determines when recovery is defined.

5 Explaining Heterogeneity

RQ3 asks which realized signal shortages are associated with damage and recovery across controlled conditions. The unit of analysis is a controlled comparison, not an individual patient. A damage comparison identifies a dataset, severity, and tail assignment; a recovery comparison additionally identifies the method and gate. Damage occurrence is not modelled because the RQ1 gate already provides that thresholded decision.

Outcomes to explain.

1. **Damage magnitude** is the signed balanced-accuracy deficit. Every controlled comparison contributes, including those showing no damage or an apparent improvement under imbalance.

2. **Recovery** is the fraction of the demonstrated deficit repaired by a mitigation method. Only gate-passing comparisons contribute because there is no stable recovery fraction without an established deficit.

Candidate explanations. Predictors are constructed without using test outcomes. Together, the four predictors in Table 4 provide one summary for each distinct dimension of the signal taxonomy.

Predictor	Question it represents	Model role
Achieved allocation severity $\log \rho$	How unequal was the realized head–tail allocation?	Joint model
Independent-evidence shortage S_{ind}	How much patient or slide support did deprived classes lose?	Joint model
Support–difficulty alignment $A = \text{corr}(\log N_c, d_c)$	Did greater support go to harder classes ($A > 0$), or did scarcity compound difficulty ($A < 0$)?	Joint model
Diversity shortage S_{div}	How much representational diversity did deprived classes lose?	Joint model

Table 4: The four RQ3 predictors. Prevalence and nominal support share one allocation-severity predictor because the fixed total T makes them two readings of the same manipulated allocation.

Both shortage measures compare the imbalanced condition with its size-matched balanced condition and include only classes allocated fewer patches under imbalance. Independent-evidence shortage measures the loss of patient or slide support. Diversity shortage measures the loss of semantic-scale log-volume on fixed Virchow2 embeddings, using a within-condition draw matched across classes in both cases and patches. This fixed, pre-mitigation measure reflects the diversity-aware method's dynamic signal without using fitted models or test outcomes.

Exactly two models are fitted: one for signed damage and one for recovery among gate-passing comparisons. Both use the same four standardized predictors and a dataset–target random intercept. Neither includes interactions nor method effects. Weak Gaussian priors and bootstrap observation errors stabilize estimation, while a leave-one-group-out check assesses whether the damage associations depend on one dataset–target group. Appendix C gives the full predictor construction and model form.

The four predictors are correlated summaries of the realized signal profile. Their coefficients therefore describe conditional associations, not separate causal effects. RQ3 is explanatory and hypothesis-generating; it does not claim predictive generalization.

RQ3 can therefore identify signal combinations that deserve follow-up, but the small number of dataset–target groups limits broader inference. The final section places this limitation alongside the benchmark's other interpretation boundaries.

6 Conclusion and Scope

This benchmark is designed to answer a bounded question: when a fixed training budget is redistributed across classes, which signal shortages appear, what damage follows, and which signal-aware interventions recover that damage? Holding the representation, partitions, model family, and each method's update policy fixed across allocation conditions makes the allocation contrast interpretable. The resulting claims concern changes in the training distribution under one frozen Virchow2 representation.

Several boundaries remain. The design does not establish how end-to-end encoder fine-tuning would respond, whether another foundation representation would produce the same deficits, or whether constructed tails reproduce biological prevalence and referral pathways. The shared total T isolates allocation by withholding some available examples. The full natural condition is therefore retained as a descriptive CE anchor, placing the controlled results alongside the complete training partition and the dataset's natural class prevalence. Because no mitigation method is fitted on that anchor, method comparisons remain confined to the size-matched controlled conditions.

Method comparisons identify methods, not universal mechanisms. The soft-F_1, soft-MCC, and cRT-stage-two arms are sampling-plus-loss hybrids, not independent sampling evidence. Exposure, time, memory, and model size are reported for the whole roster.

Evidence also depends on independent support, not merely patch count: the discrimination and calibration gates are evaluated on the case-macro estimator over each class's pooled cell across split appearances (Appendix B), and inference is withheld as descriptive wherever that pooled cell's Kish effective support, single-unit dominance, or contributing-unit count fails the prespecified threshold. Calibration has a separate boundary: it is conditional on the natural held-out prevalence and cannot be transported to another deployment population without an explicit prior-shift analysis.

Within these boundaries, the three research questions form a deliberate sequence. RQ1 establishes the signal profile and damage created by controlled allocation. RQ2 asks which interventions recover performance only after that damage is established. RQ3 then relates variation in damage and recovery to the realized shortages without treating those associations as causal decomposition. Together, this sequence turns class imbalance from a single prevalence label into a controlled analysis of what evidence is missing, what that shortage changes, and which signal a mitigation method should use.

A Construction and Support Specification

This appendix records dataset eligibility, hierarchical sampling, support floors, allocation, and seed rules. These details implement the controlled contrast introduced in Section 3; they do not add further estimands.

Locked quantity	Value	Spec.
Representation	Frozen Virchow2 features	A
Patch classifier	Two-layer MLP, 512 hidden units	A
Example budget	$E = 30T$ example presentations	D
Learning-rate candidate range	Nine values, 3×10^{-6} to 3×10^{-2}	D.2
Support floor per class	≥ 10 partitioning units and ≥ 20 slides	A.2
Contribution caps	$\leq 10\%$ per partitioning unit; $\leq 5\%$ per slide	A.2
Severity targets	$\rho_{\text{target}} \in \{1, 10, 100\}$	A
Severity tolerance	Achieved $\rho \in [9, 11]$ and $[90, 110]$	A.2
Tail assignments	Native, difficulty-aligned, difficulty-reversed	A
Data partitions	Three locked, patient- or case-disjoint	A
Initialization seeds	Two tuning; five disjoint confirmation	D
Gate aggregation unit	Case-macro (patch-micro secondary)	4
Evidence cell	Pooled across a case's split appearances	B
Deficit gate thresholds	$D_{\text{BA}} \geq 0.01443$; $D_{\text{cal}} \geq 0.18237$ nats; both with CI excluding 0	4.3

Table 5: Design and training quantities locked before the first definitive fit, with the appendix section that specifies each. The allocation of the shared training support T among classes is the only directly manipulated quantity (Section 3).

A.1 Eligibility and representation

Before partitions are derived, the source manifest is restricted to eligible examples with fixed features. A sample is excluded if it lacks a valid target or partitioning-unit identifier, yields no patch under the dataset-specific tissue rule, lacks an annotation required for its patch target, or lacks readable fixed features. Each remaining partitioning unit is then assigned to exactly one training, validation, or test partition, and all slides from that unit follow its assignment. Eligibility is fixed once rather than reapplied by imbalance condition.

TCGA-UT further excludes two of its 32 annotated cancer types from the eligible target set, for two independent reasons. Diffuse Large B-cell Lymphoma contributes 20 participants and 20 slides in every locked split, one slide per participant, so the 10% partitioning-unit and 5% slide patch contribution caps (Section A.2) admit essentially no count above the definitive floor: the class's set of cap-feasible patch counts covers only a few percent of the range between that floor and its native support, with the first infeasible count one unit above the floor. No shared total is therefore both tolerance-feasible and cap-feasible for this class, independent of severity or tail assignment. Cholangiocarcinoma is excluded on a separate, evidence-side ground: its held-out participant count cannot support the split-unit-clustered inference the confirmatory analysis requires. The remaining 30 cancer types are eligible and satisfy the tolerance search below.

All eligible patches enter the pool before condition-specific subsampling. The common classifier is a two-layer MLP with a 512-unit hidden representation, ReLU activation, dropout, and a linear class output. CFAL replaces the linear output with a prototype-based affinity head; OKO adds a training-only auxiliary head discarded at evaluation. Dataset provenance and complete preprocessing definitions remain in the companion dataset report (Queisler, 2026a).

A.2 Independent evidence and support floors

The independent-support rule of Section 3 is realized as a fixed per-class partitioning-unit pool and, within it, a fixed slide pool for the concentrated conditions. The paired spread conditions use the same seeded ordering but extend each class pool to its eligible partitioning-unit inventory while preserving the nominal patch allocation. Thus the independent-support contrast is deliberately between pools at fixed nominal allocation; the concentrated pool is a patient-prefix subset of the corresponding spread pool. Table 6 states the selection order and caps.

Selection order	Contribution caps	What imbalance changes
Partitioning units, then slides within units, then patches; deterministic round-robin before any unit is revisited	$\leq 10\%$ of a class's patch allocation per partitioning unit; $\leq 5\%$ per slide	Nominal allocation, or independent support through the paired concentrated and spread pools

Table 6: Hierarchical selection and contribution caps. Every sample list reports per-class partitioning-unit, slide, and patch counts, maximum contributions, and the fraction of the eligible independent pool retained.

Within each pool arm, the condition with the largest patch allocation for a class covers its fixed pool in full, so no unit in that arm goes unused by every condition together. A smaller condition for the same class, in particular a severe tail count well below that maximum, may leave some pool units untouched; this is the intended shape of an imbalanced allocation, not a violation of the fixed-pool construction.

The smallest defensible tail support is estimated before the definitive comparison with a *support pilot*. These balanced CE runs ask when additional independent training units cease to materially improve natural-validation performance; they do not tune a hyperparameter or contribute observations to the definitive analysis. One pilot unit is a partitioning unit, or a slide where partitioning unit and slide coincide. Each selected unit contributes the same fixed quota q_{patch} of patches, distributed as evenly as possible over its eligible slides. The quota is the largest value the patient and slide contribution caps admit at every candidate level and every construction ordering, held fixed so increasing a pilot level adds independent units rather than patches per unit. A caps-admissible quota is necessary but not sufficient: a unit cannot supply more patches than it holds, so a unit is eligible for a candidate level only if its own inventory meets the candidate quota, and the quota at a given level is set from that level's order statistic of per-unit inventory over the eligible units rather than from the scarcest unit in one seeded ordering. A candidate level that no quota can satisfy is dropped from the curve rather than reusing a looser quota from a larger level.

Nested pilots compare $10, 15, 20, 30$, and 50 units per class, or all available units when the scarcest class has fewer. They are repeated for three independently seeded *construction orderings*. An ordering is a fixed sequence of eligible units; its prefix of length m is simply the first m units in that sequence. Thus the 15-unit pilot retains the same first 10 units as the 10-unit pilot and adds the next five. At lower levels of the hierarchy, every partitioning unit has a fixed slide order and every slide has a fixed patch order. Figure 6 illustrates this nesting. Because every larger pilot level extends the preceding prefix, adjacent levels are paired within an ordering and evaluated on the same natural-validation cohort. The smallest level whose next increment changes mean balanced accuracy by less than 0.01 and every class recall by less than 0.02 in all three orderings is the *stability floor*; if none qualifies, the largest candidate is used.[1]

[1] No pilot level is exempt from the caps. A level is dropped from the curve when no quota satisfies both caps for every construction seed—e.g., the 5% slide cap needs at least 20 contributing slides, so a class whose partitioning units mostly contribute a single slide can leave levels below 20 infeasible. Dropped levels are recorded in the frozen pilot report.

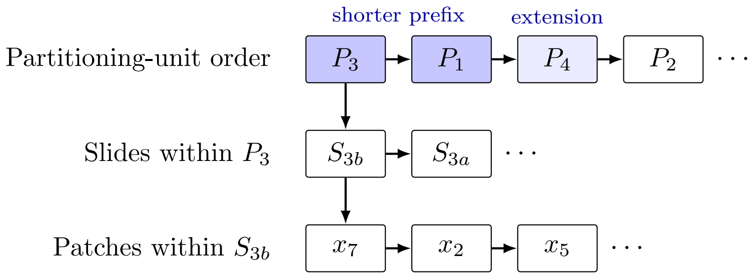

Figure 6: Hierarchy behind one construction ordering. A seed fixes the partitioning-unit sequence and the local slide and patch sequences. The darker unit boxes form a shorter prefix; the lighter third box is added, without replacing earlier units, when the prefix is extended.

The definitive minimum is the larger of this empirical stability floor and a prespecified design floor. Each class must contain at least 10 partitioning units and 20 slides, or 20 slides when partitioning unit and slide coincide, and any class-aware batch or set must contain at least two distinct same-class units. The 10-unit and 20-slide values are not estimated performance thresholds: they are the smallest counts compatible with the 10% partitioning-unit and 5% slide contribution caps, respectively. A requested severity is reduced to the largest allocation satisfying both floors; a dataset is excluded if even its balanced condition fails them. Pilot quotas, curves, seeds, and resulting floors are frozen and reported with the definitive training sets.

Within these floors, one shared total T is fixed for the balanced, moderate, and severe conditions of a split. Construction scores every integer total between the independent-support floor and the largest total any locked tail assignment's head class can support at $\rho_{\text{target}} = 100$, keeping only totals whose balanced allocation and whose every locked assignment's moderate and severe allocation satisfy the patch or slide contribution caps. Among these cap-feasible totals, the *tolerance-feasible* set is those whose achieved moderate ratio falls in $[9, 11]$ and whose achieved severe ratio falls in $[90, 110]$ simultaneously, for every locked assignment. The largest tolerance-feasible total is checked against the true fixed-pool construction — the same sampling procedure the definitive training sets use — starting from the largest candidate and moving to the next largest until one is confirmed; that confirmed total is T. The main-text tolerance statement describes this preferred construction and all realized conditions reported in Table 2. A dataset–regime with no tolerance-feasible total instead activates the explicitly labelled contingency below.

Before assigning support ranks, the largest balanced pilot level is used to measure class difficulty. Each of the three pilot constructions supplies one hierarchically matched draw in the sense of the companion methods report (Queisler, 2026b): every class contributes the same number of partitioning units and every selected unit the same number of patches, so the probe sees equal independent support in every class rather than only equal patch counts. On each draw a class-balanced linear probe makes five-fold out-of-fold predictions with case/patient-disjoint `StratifiedGroupKFold` partitions. Class difficulty is $d_c = 1 - \overline{\text{recall}}_c$, where recall is averaged across the three matched draws; five-neighbour kNN recall and the unmatched, patch-balanced-only probe error are retained as sensitivity evidence, the latter because a class ordering that changes between the matched and unmatched scores marks a class whose apparent difficulty is partly scarcity. Pilot hashes, folds, classwise scores, and the easiest-to-hardest ranking are signed. Equal scores retain native class order.

A.3 Severity profiles and tail assignments

A binary task needs only a head and a tail count, but a multiclass task also needs a rule for the classes between those extremes. The exponential construction places the K class supports at equal intervals on a logarithmic scale. For a locked order from head ($i = 0$) to tail ($i = K - 1$), it assigns weights

$$w_i(\rho_{\text{target}}) = \rho_{\text{target}}^{-i/(K-1)}, \qquad n_i^* = T \frac{w_i(\rho_{\text{target}})}{\sum_{j=0}^{K-1} w_j(\rho_{\text{target}})}, \quad i = 0, \ldots, K-1. \tag{4}$$

so the head-to-tail ratio is exactly ρ_{target} before integer rounding. For example, with four classes and $\rho_{\text{target}} = 100$, the unnormalized weights are approximately $1, 0.215, 0.046$, and 0.01; the two body classes lie geometrically between the head and tail instead of being assigned ad hoc counts. Binary targets use $n_{\text{head}}^*/n_{\text{tail}}^* = \rho_{\text{target}}$ directly. The same exponential rule is applied to every multiclass dataset–regime, so a given ρ_{target} denotes the same allocation shape in each of them.

The values n_i^* are real-valued targets, whereas a training set must contain whole patches or slides. *Constrained integer allocation* converts them into usable class counts. Each target is rounded to the nearest integer and clipped to the class's support floor and available unique support. If the rounded counts do not sum to T, units are added to or removed from the classes with the largest rounding residuals until the total is exact. The resulting integers n_i therefore satisfy $\sum_i n_i = T$ and every class-specific lower and upper bound while remaining close to the exponential targets under this deterministic adjustment.

Counts determine how many units a class contributes; construction orderings determine which units they are. One master ordering is created per class at each hierarchy level. Every condition selects a deterministic prefix, or a partitioning-unit-round-robin extension of a prefix, from these same orderings: all fixed-pool partitioning units and slides contribute before additional patches are allocated.

If no total satisfies both tolerance bands, the same contingency is applied to every dataset and regime: use the largest jointly attainable moderate and severe ratios, then retain the largest total attaining both maxima. The condition is labelled *off-target*; its requested ratio, achieved ratio, limiting class, realized counts, and binding support floor are reported, and it is interpreted by its achieved rather than nominal severity. Validation or test support is never reduced to manufacture feasibility. This contingency is especially relevant to high-K tasks, where an exponential profile must allocate support to many intermediate classes as well as the tail; it is not a dataset-specific exception. A stable lower-severity condition is preferred to a nominal $\rho = 100$ condition supported by only a few independent units.

The fallback has a limit: a moderate or severe condition whose realized allocation is indistinguishable from the balanced condition—achieved ratio within a small tolerance of 1, or realized counts identical to the balanced training set—is not a lower-severity instance of the intended contrast but a null one, since the imbalance intervention did not occur. Such a condition is reported as degenerate rather than fitted and averaged with genuine moderate or severe conditions from other splits or assignments, and it is excluded from the imbalance deficit and recovery estimands in Section 4.3. Its requested ratio, achieved ratio, limiting class, and binding support floor are recorded alongside the exclusion, using the same fields every condition already reports.

The profile determines support by rank; the tail assignment maps semantic classes to those ranks. Native order is the clinical, ordinal, or natural-support order of the target; aligned and reversed order the classes by the difficulty scores d_c measured above. Binary tasks retain native and reversed orientations only. A difficulty mapping that duplicates an earlier mapping in any locked split is omitted across that dataset–regime with its reason recorded. These three assignments separate interpretable mechanisms but do not exhaust all possible class-support mappings. Figure 3 illustrates the design.

The random quantities are separated into five seed families so that changing one source of variation does not silently change another. Table 7 defines their roles.

Seed family	Determines	Prespecified use
Data partition	Membership of the training, validation, and test cohorts	Three locked patient- or case-disjoint splits
Tail assignment	Mapping of semantic classes to head–tail support ranks	Native, difficulty-aligned, and difficulty-reversed mappings from signed pilot evidence
Construction	Partitioning-unit, slide, and patch orderings and the selected evidence	Three pilot orderings and a disjoint fourth ordering for definitive training sets
Initialization	Parameter initialization and optimization randomness	Two tuning seeds and five disjoint confirmation seeds
Resampling	Bootstrap draws and permutation swaps	Uncertainty intervals and hypothesis tests only

Table 7: Seed families and their non-overlapping roles in the benchmark.

No seed is chosen based on test performance. The size-matched balanced training set and its trained CE baseline are shared by all tail assignments within a split.[2] The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

B Endpoint and Inference Specification

This appendix defines secondary endpoints, method-specific score semantics, resampling, diagnostics, and confirmatory testing. Gates and recovery retain the main-text definitions in Section 4.3.

B.1 Probability quality and calibration

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \dots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0,1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i.

Table 8 fixes which score variant enters each endpoint. Ordinary CE, post-hoc logit adjustment, and train-time logit adjustment have an explicit target-prior correction defined in the companion methods report, using the natural-validation prior for the corresponding test split. No model-free target-prior correction is defined for the other methods; their normalized raw scores are therefore retained and labelled as such rather than described as target-prior-corrected probabilities.

This distinction limits interpretation. NLL and Brier score are proper scoring rules for probability forecasts. For methods without an explicit prior correction, the same equations instead assess the method's normalized raw scores on the natural held-out cohort, without asserting that those scores estimate target-prevalence probabilities. In particular, CFAL's normalized affinities are ranking scores rather than calibrated probabilities, as specified in the companion methods report; its NLL, Brier score, and ECE are therefore diagnostic score summaries and do not support a calibrated-probability claim. The CE-only deficit gate remains target-prior-corrected, while method-specific tail-NLL recovery is reported with the score variant identified in Table 8.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\mathrm{NLL} = -\frac{1}{N} \sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{5}$$

[2] At $\rho_{\text{target}} = 1$, every class receives equal support. Permuting semantic classes across support ranks therefore leaves the allocation unchanged, so tail assignment is not a meaningful factor for the balanced condition.

Endpoint or output	Score variant
Balanced accuracy and all other discrimination endpoints	Balanced-decision logits: $z^{\mathrm{CE}} - \tau \log \widehat{\pi}$ for post-hoc logit adjustment; raw model logits for train-time logit adjustment and every other method.
Tail-NLL deficit gate	Target-prior-corrected CE probabilities before temperature scaling.
Tail-NLL recovery	CE and both logit-adjustment variants: target-prior-corrected probabilities; all other methods: normalized raw scores.
Reported NLL, Brier score, ECE, and reliability diagrams	The same method-specific variant as tail-NLL recovery; uncorrected outputs are explicitly labelled raw-score results.
Temperature-scaled calibration outputs	The applicable logits or raw scores divided by one positive temperature fitted on natural-validation NLL and applied unchanged to test; these secondary outputs do not enter either recovery gate.

Table 8: Canonical mapping from evaluation endpoints to score variants.

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i,y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\mathrm{NLL}_c = -\frac{1}{N_c} \sum_{i:y_i=c} \log \hat{p}_{ic}, \qquad \mathrm{NLL}_{\mathrm{macro}} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{NLL}_c, \tag{6}$$

where N_c is the number of held-out predictions from class c. Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2 . \tag{7}$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left| \frac{1}{|I_b|} \sum_{i \in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|} \sum_{i \in I_b} q_i \right|, \tag{8}$$

where I_b contains the predictions assigned to bin b. ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and partitioning-unit weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. The output semantics and prior-correction formulas of individual methods are defined in the companion methods report (Queisler, 2026b).

B.2 Uncertainty from resampling partitioning units

The resampled partitioning unit is the patient or participant where one is available, and the biopsy/WSI for PANDA and CAMELYON16. Each replicate reweights the observed partitioning units rather than deterministically repeating each one exactly once, and the complete statistic is recomputed from those weights.

Bootstrap patient-support check. Before model training, 10,000 bootstrap replicates are constructed using only partitioning-unit identifiers, split membership, and class labels. No predictions, performance values, or method identities are used; these replicates test whether the planned resampling scheme is valid before results can influence it. The count of 10,000 is a prespecified Monte Carlo budget. It places about 250 replicates in each 2.5% tail of a percentile interval, reducing simulation noise in the interval endpoints while remaining inexpensive before model fitting.

Each partitioning unit draws an independent, continuous bootstrap weight in every replicate: a Bayesian bootstrap (Rubin, 1981) in which the n observed units' weights are n times a draw from a symmetric Dirichlet$(1, \ldots, 1)$ distribution over the n-simplex. This draw is almost surely strictly positive in every coordinate, so every unit contributes to every replicate; no split–class cell, including one contributed by a unit whose observed split-by-class pattern is unique among the cohort, can be emptied by an unlucky draw. A unit's observed contribution pattern—how many examples it supplies to each class in each test-split appearance—is still recorded and grouped into strata, but only to report per-stratum diagnostics; it no longer constrains the weight draw itself. A selected unit's weight is applied to all of its slides and patches in every split. Figure 7 summarizes the workflow.

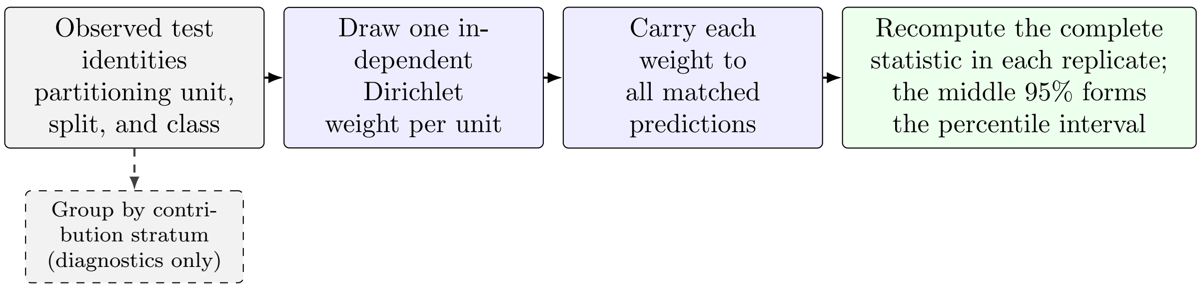

Figure 7: Bayesian bootstrap workflow clustered by partitioning unit. The bootstrap patient-support check executes the identity, weight-draw, and stratum-diagnostic steps before predictions exist. After model fitting, the same paired per-unit weights are applied to every matched run.

The patient-support check verifies that every original split–class cell remains represented, every split-level metric is computable, one partitioning unit receives the same weight across all of its split appearances, and every unit's resampled weight actually varies across replicates. Failure of any check stops the analysis rather than causing a replicate to be silently discarded.

The independent per-unit weight draw can still produce a replicate dominated by a few heavily weighted units. If partitioning unit i has class-specific bootstrap weight v_i, the *Kish effective partitioning-unit count*

$$n_{\text{Kish}} = \frac{\left(\sum_i v_i\right)^2}{\sum_i v_i^2}. \tag{9}$$

translates unequal case weights into the equivalent number of equally weighted partitioning units. It decreases as weight concentrates. The patient-support check also records, pooled across the union of a class's split appearances, its number of contributing units and its largest unit-weight share — the cell the shared per-unit weight draw actually resamples against, since one unit's weight applies to every split appearance it makes at once. Table 9 lists the three concentration checks, each evaluated on that pooled cell; failing any one designates inference for that dataset–regime descriptive before training. This is separate from the split-level representation check described above, which still requires every original split–class cell to remain represented.

Diagnostic	Threshold for descriptive-only designation
Effective partitioning-unit count	2.5th percentile of n_{Kish} across replicates below five
Single-unit dominance	One unit supplies more than 50% of a class's weight in more than 5% of replicates
Cell support	A class's pooled cell (union of its split appearances) has fewer than five contributing units

Table 9: Concentration criteria for the bootstrap patient-support check, each evaluated on a class's cell pooled across the union of its split appearances — the cell the shared per-unit weight draw resamples against. All three are evaluated before predictions exist; the designation is recorded rather than used to discard replicates.

A percentile rather than the minimum is used because the minimum across replicates is an extreme order statistic: its value falls as the replicate count rises, so a floor placed on it would encode the resampling budget rather than the support the cell actually carries. The diagnostic report and resampling seed are frozen with the analysis plan.

Intervals after model fitting. The three partitioning-unit-disjoint data splits are fixed design repetitions and are averaged with equal weight; they are not treated as three independent cohorts. Tail assignments remain separate unless an assignment-averaged estimand is explicitly reported. Within each case-bootstrap replicate, the five matched confirmation-seed indices are also resampled as paired blocks and averaged before the splits are combined. This captures algorithmic variation without pretending that five model initializations create five additional case cohorts.

Balanced CE, imbalanced CE, and each mitigation method receive the same partitioning-unit weights and confirmation-seed draws. The complete deficit and recovery fraction, including numerator and denominator, is recomputed in each replicate. The reported 95% percentile interval spans the 2.5th to 97.5th percentiles of the resulting distribution. Patches are never bootstrapped independently.

B.3 Paired testing and multiplicity control

Hypothesis tests provide a supporting check for the prespecified primary methods. On the discrimination axis, the tested effect is the balanced-accuracy improvement over imbalanced CE. On the calibration axis, it is the reduction in tail-group macro NLL, oriented so that positive values indicate improvement. Recovery fractions are normalized effect summaries, not test statistics.

The raw two-sided p-value comes from a paired partitioning-unit-block permutation test. This test represents the null hypothesis of no method difference by repeatedly swapping the method and imbalanced-CE labels within a partitioning unit. The same swap is used for all of that unit's patches, split appearances, and corresponding confirmation-seed indices, preserving the paired design. All possible swaps are enumerated when feasible. Otherwise, 100,000 random swaps are used with a plus-one correction; this gives a minimum attainable Monte Carlo p-value of approximately 10^{-5} and avoids reporting a simulated value of zero.

A *test family* is the set of related hypotheses interpreted together. Testing many hypotheses increases the chance that at least one small p-value occurs by chance. Holm adjustment controls this family-wise error by ordering the raw p-values from smallest to largest, applying the strongest correction to the smallest value, and then proceeding stepwise with less stringent corrections. It provides this control without assuming that the tests are independent.

The confirmatory family holds one method per signal: weighted CE (prevalence), class-balanced CE (nominal support), independent-support weighted CE (independent support), pilot-difficulty weighted CE (difficulty), and semantic-scale weighted CE (diversity). All five share one intervention level, a mean-one class weight on unmodified CE (Section 4.5), so their effect estimates can be compared without changing the intervention mechanism. Within one dataset, the family includes every gate-passing combination of primary method, gate, severity, and tail assignment; severities are not split into separate families. Holm adjustment is applied across the testable method-versus-imbalanced-CE comparisons, while gated-out comparisons are recorded as not tested. Every other roster member remains exploratory and outside the confirmatory correction.

These tests establish whether each primary method improves on imbalanced CE, but on their own they cannot answer the matching part of RQ2: significance for one method and non-significance for another is not evidence that the methods differ. A direct between-method contrast is therefore prespecified, together with the rule that decides which method is matched.

Prespecified matching rule. Every imbalanced comparison unit receives one *dominant shortage* label, computed before any recovery outcome is inspected and from pre-outcome quantities only. Four shortage scores summarize its realized signal profile over the deprived class set $\mathcal{T}_j$ of Appendix C: the nominal-allocation shortage

$$S_{\text{nom},j} = \frac{1}{|\mathcal{T}_j|} \sum_{c \in \mathcal{T}_j} \log \frac{N_c^{(\text{bal})}}{N_{cj}^{(\text{imb})}}, \tag{10}$$

which prevalence and nominal support share because the fixed total T makes them two readings of the same allocation; the independent-evidence shortage $S_{\text{ind},j}$ of Equation (11); the difficulty-compounding score $S_{\text{dif},j} = -A_j$, positive when scarce support fell on harder classes; and the diversity shortage $S_{\text{div},j}$ of Equation (13). Each score is standardized across all controlled comparisons, and the dominant shortage is the largest standardized score. The *matched* members are the confirmatory methods reading that signal: weighted CE and class-balanced CE when S_{nom} dominates, independent-support weighted CE for S_{ind}, pilot-difficulty weighted CE for S_{dif}, and semantic-scale weighted CE for S_{div}. A unit whose two largest standardized scores lie within 0.25 standard deviations of each other is labelled *ambiguous*: it carries no matched label, contributes no direct contrast, and enters the descriptive comparison only. Standardization is a prespecified ordering device, not a claim that unlike shortages are commensurable on a common natural scale.

Direct between-method contrast. For each gate-passing, non-ambiguous comparison unit, the contrast is the paired difference on the gated endpoint between the matched method and the mean of the unmatched confirmatory members, evaluated on the same held-out cases, partitioning-unit weights, and confirmation-seed draws, and tested with the same partitioning-unit-block permutation procedure. When S_{nom} dominates and two members are matched, their mean forms the matched side. Within one dataset the confirmatory family therefore contains, per gate-passing comparison unit, the five method-versus-imbalanced-CE tests and this one matched-versus-unmatched test, all adjusted jointly by Holm. Pairwise matched-versus-single-unmatched differences and the five raw effects and recovery fractions are reported alongside the pre-outcome signal profile from RQ1 with confidence intervals, but remain exploratory.

This family and its matching rule replace the intervention-based family originally specified here, whose rationale assumed a second prediction regime that the benchmark no longer carries. Both changes were fixed before any recovery outcome was inspected. The signal-based family is confirmatory only under that timing and must not be revised again after recovery outcomes are read.

C RQ3 Model Specification

This appendix fixes the four signal predictors and two association models used for RQ3. No predictor uses test labels, deficit estimates, recovery estimates, or fitted mitigation-model outputs.

Balanced-to-imbalanced shortages. For controlled comparison j, let $\mathcal{T}_j = \{c : N_{cj}^{(\mathrm{imb})} < N_c^{(\mathrm{bal})}\}$ contain the semantic classes receiving fewer nominal patches than in the shared balanced condition. The independent-evidence shortage is

$$S_{\mathrm{ind},j} = \frac{1}{|\mathcal{T}_j|} \sum_{c \in \mathcal{T}_j} \log \frac{G_c^{(\mathrm{bal})}}{G_{cj}^{(\mathrm{imb})}}, \tag{11}$$

where G_c counts contributing patients or slides. Positive values mean that nominally deprived classes also lost independent evidence.

Representational diversity is measured on frozen Virchow2 features using the same within-class semantic volume as semantic-scale weighted CE,

$$S'_c = \tfrac{1}{2} \log_2 \det\left(I + \frac{d}{M\varepsilon^2} Z_c Z_c^\top \right). \tag{12}$$

Within each condition, the deterministic draw contains the same number of patients and patches per class, and pooled isotropic scaling fixes $\varepsilon = 1$. With the numerical floor $\varepsilon_S = 10^{-3}$, diversity shortage is

$$S_{\mathrm{div},j} = \frac{1}{|\mathcal{T}_j|} \sum_{c \in \mathcal{T}_j} \log \frac{\max(S_c'^{(\mathrm{bal})}, \varepsilon_S)}{\max(S_{cj}'^{(\mathrm{imb})}, \varepsilon_S)}. \tag{13}$$

Positive values indicate lower observed feature volume among nominally deprived classes. If no class is deprived, both shortage contrasts are defined as zero.

Two association models. For either signed balanced-accuracy damage or gated recovery, the linear predictor is

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\mathrm{ind}} S_{\mathrm{ind},j} + \beta_A A_j + \beta_{\mathrm{div}} S_{\mathrm{div},j}, \tag{14}$$

where $u_{g[j]}$ is the random intercept for the dataset–target group and $A_j = \mathrm{corr}(\log N_{cj}, d_c)$. Every predictor is standardized before fitting. The damage model includes every controlled comparison and incorporates its bootstrap standard error; the recovery model includes only comparisons passing a prespecified RQ1 deficit gate and incorporates the recovery-ratio bootstrap standard error. No occurrence model, interaction, method effect, alternative predictor model, or diagnostic model is fitted.

Parameters are estimated by maximum a posteriori (MAP) fitting with zero-centered unit-scale Gaussian penalties on standardized slopes and log-scale parameters. A leave-one-dataset–target-group-out check refits the damage model while withholding one complete group at a time. This is only a coarse stability check, not evidence of predictive generalization. Model definitions, predictor roles, and priors are frozen before recovery outcomes are inspected, and all RQ3 conclusions remain hypothesis-generating.

D Training, Tuning, and Execution Specification

This appendix records exposure budgets, validation search, replication, and resource accounting.

D.1 Exposure budgets and checkpoint selection

With B the locked batch size and T the common controlled support, every fitted method receives the frozen example-presentation budget

$$E = 30T. \tag{15}$$

Its update count is $U_m = \lceil E/e_m \rceil$, where e_m is that method's examples per update. Epochs are neither a stopping rule nor a comparison unit.

Method	Examples per update e_m
Single-stage default	B
OKO	$B(k+2)$
MDE	$2B$
cRT	$1.2B$ across both stages
Natural anchor (CE only)	$E = 30T_{\text{natural}}$; descriptive

Table 10: Exposure accounting. The frozen E is common; U_m is derived per method and selected configuration. Post-hoc logit adjustment performs no gradient update and reuses CE.

Within a run, the reported checkpoint is fixed by three rules:

1. Validation is stored every $\max(1, \text{round}(U_m/170))$ updates, giving about 170 checkpoints per fit.
2. The selected checkpoint maximizes natural-validation balanced accuracy; macro F1 and then lower NLL are deterministic tie-breakers.
3. Test predictions are produced once from that locked checkpoint.

D.2 Control grids and adaptive validation search

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity, and the search adapts within a prespecified candidate range rather than committing to a single fixed grid. Every trainable method draws its learning rate from the common candidate range

$$\eta \in \{3\times10^{-6}, 1\times10^{-5}, 3\times10^{-5}, 1\times10^{-4}, 3\times10^{-4}, 1\times10^{-3}, 3\times10^{-3}, 1\times10^{-2}, 3\times10^{-2}\}. \tag{16}$$

Table 11 lists each method's imbalance-specific control. Focal loss, CE + soft F1, and CE + soft MCC use the same adaptive window as the learning rate; the other controls are bounded four-value grids. The prior outer learning-rate winners, 1×10^{-5} and 1×10^{-2}, are interior to this rerun's candidate range.

Matched counting-unit diagnostic. Class-balanced and independent-support weighted CE both weight by $1/E_c(\beta)$, the inverse effective number from the same saturating transform, evaluated at nominal count n_c for the former and independent count G_c for the latter, as defined in the companion methods report (Queisler, 2026b). Because β sets both a saturation scale and, jointly with the counting unit, the realized weight shape, tuning β separately for each method confounds the counting unit with the selected saturation scale. The tuning grid therefore includes a *matched-beta* arm: independent-support weighted CE is confirmed a second time at class-balanced CE's tuned β, with its own tuned learning rate. Only this matched-β comparison attributes a difference to the counting unit alone; it is a diagnostic arm reported alongside, not in place of, the five-family contrast, and is not itself one of the five confirmatory members.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0, 0.5, 1, 1.5, 2, 4, 8\}$ candidate range; active window starts at $\{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0, 0.25, 1, 4, 16, 64\}$ candidate range; active window starts at $\{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
Class-balanced CE	$\beta \in \{0.9, 0.99, 0.999, 0.9999\}$; saturation points $1/(1-\beta) \in \{10, 100, 1000, 10000\}$, matched to n_c's range
Independent-support weighted CE	$\beta \in \{0.8, 0.95, 0.99, 0.999\}$; saturation points $\{5, 20, 100, 1000\}$, scaled down to G_c's range, plus a matched-β arm at class-balanced CE's tuned β
Pilot-difficulty weighted CE	$\tau \in \{0.25, 0.5, 0.75, 1\}$
Semantic-scale weighted CE	$\tau \in \{0.25, 0.5, 0.75, 1\}$; reweighting begins at reference pass 6 of 30

Table 11: Prespecified method-specific controls, crossed with the active window of the learning-rate candidate range in Equation (16). Focal loss and the two soft-label hybrids search their listed candidate set through an adaptive four-value window; every other row is a fixed four-value grid. cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately. The class-balanced and independent-support β grids are matched saturation points on the effective-number transform, scaled to each method's own counting unit as defined in the companion methods report (Queisler, 2026b).

Method-specific implementation deviations (CFAL's fixed constants, semantic-scale's warmup and degenerate-draw fallback, focal's alpha-balancing) are documented in the companion methods report (Queisler, 2026b). Table 3 already reflects focal's resulting family reassignment.

Adaptive window search, run once per method, dataset, and imbalance severity.

1. Initialize each active window to four adjacent candidate values: $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}$ for the learning rate, and the start listed in Table 11 for an adaptive strength axis.
2. Fit the active learning-rate window crossed with the method's strength window or fixed grid: at most 16 candidates per round (four for CE/cRT; one for post-hoc adjustment). Earlier candidates are reused.
3. If an axis's winning value is interior to its window, record the axis as *resolved* and stop searching it.
4. If the winner is an edge value and the candidate range extends further in that direction, shift the window one position toward that edge and return to step 2. The three overlapping axis values are reused. Only configurations introduced by the new row or column are fitted: four when one axis shifts and seven when both shift together.
5. If the winner is an edge value at the candidate range's boundary, retain the boundary value and record the axis as *tuning-limited*.

Round bounds. At most three rounds on the learning-rate axis, three on focal loss's γ axis, and two on the soft-F1 and soft-MCC strength axis, each bounded by the number of available candidates. A resolved or tuning-limited axis is never retrained in a later round.

For weighted CE, balanced sampling, focal loss, CE + soft F1, and CE + soft MCC, an imbalance-specific strength of zero is mathematically identical to plain CE at the same learning rate: it applies no class reweighting or sampling adjustment, reduces the focal term to the plain cross-entropy loss, or drops the auxiliary term entirely. That value is therefore excluded from the searched window, and its metrics are copied from CE's already-fitted same-learning-rate run rather than retrained, giving each of these five methods one additional comparison point at no extra training cost.

Four method families deviate from the search above. CE and methods without a meaningful imbalance-specific control search the learning-rate candidate range alone. Train-time logit adjustment is tuned jointly over τ and learning rate, with τ held to its fixed grid. Post-hoc logit adjustment inherits the selected CE model and tunes only τ, because it performs no gradient optimization. cRT inherits the selected CE configuration for stage one and selects its stage-two classifier learning rate separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

D.3 Replication, locking, and records

Selection and replication follow prespecified units rather than per-run judgement:

- **Tuning seeds.** Every candidate is fitted with the same two tuning initialization seeds for each prespecified data split and tail assignment.
- **Selection unit.** One method, dataset, and imbalance severity. Natural-validation balanced accuracy is averaged across seeds, splits, and tail assignments; macro F1 then lower NLL break ties. NLL is not separately optimized, and pooling assignments can dilute assignment-specific optima.
- **Locking.** The chosen configuration is locked before any of the five disjoint confirmation initialization seeds is fitted and before test evaluation. Tuning runs never enter confirmation effect estimates.
- **Confirmation.** Each locked data split and tail assignment is repeated with the five confirmation seeds. Exactly three partitioning-unit-disjoint split seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis.
- **Matching.** The two runs of a controlled contrast (Appendix E) share data split, tail assignment, confirmation-seed index, and held-out cases, so the contrast does not mix the intended change with a different cohort or replication.
- **Failures.** The unit of replication and all missing or failed runs are reported. A failed run is never silently replaced using a test-informed seed.

Split counts, support floors, condition training sets, seeds, method grids, and exposure budgets are frozen in the signed analysis plan before the first definitive fit. Every candidate, window shift, and resolved-or-tuning-limited outcome is recorded before confirmation. No test or confirmation result informs a shift.

For every realized condition, the experiment record contains

- dataset version and eligibility rules;
- partitioning-unit, slide, and patch identifiers by partition;
- requested and achieved T and ρ, and the class assignment;
- all seed families and their tuning, confirmation, pilot, or definitive roles;
- pilot patch quota and bootstrap patient-support diagnostics;
- model and optimizer configuration, candidate grid, and selected checkpoint;

- environment identifier and test-prediction hash.

Deviations are dated and justified before the affected test labels are accessed.

Computational cost is separated into validation search and locked-configuration fitting, and reported as wall-clock training time, accelerator-hours, and peak accelerator memory; total and trainable parameter counts; and effective passes through unique training examples. Costs include every method-specific stage, including both cRT stages, but exclude the one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

E Terminology

Natural anchor. The full eligible natural training partition, fitted with the CE reference alone. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M.

Imbalance severity. The requested training allocation summarized by ρ; achieved severity is computed from realized integer counts.

Pilot level. A candidate number of independent training units per class used in the balanced CE support-stability study.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Comparison unit. One dataset, imbalance severity, and tail assignment evaluated together.

Deficit gate. A prespecified threshold rule that determines whether an observed discrimination or calibration deficit is large and precise enough for recovery analysis.

Imbalance deficit. The paired, sign-oriented difference at common T and on the same held-out cases: balanced CE minus imbalanced CE for higher-is-better endpoints, and imbalanced CE minus balanced CE for lower-is-better endpoints. Positive values denote harm from imbalance.

Recovery. A method's improvement over imbalanced CE divided by the evidenced imbalance deficit.

Kish effective partitioning-unit count. The equally weighted partitioning-unit count with the same weight concentration as one bootstrap replicate.

Signal profile. The measured description of a realized condition on the five signal dimensions: class prevalence, nominal patch support, independent patient or slide support, class difficulty, and representational diversity.

Independent-evidence shortage. The mean log loss of contributing patients or slides in the nominally deprived classes, relative to the size-matched balanced condition.

Diversity shortage. The mean log loss of within-class semantic volume on frozen features in the nominally deprived classes, relative to the size-matched balanced condition.

Support–difficulty alignment. The correlation between log allocated support and pilot difficulty across classes; negative values mean scarce support fell on harder classes.

Dominant shortage. The largest standardized shortage score of a comparison unit under the prespecified matching rule; it names the confirmatory members counted as matched, or labels the unit ambiguous when the two leading scores are too close.

Confirmatory signal family. The five weighted-CE methods, one per signal, together with the matched-versus-unmatched contrasts, whose tests are Holm-adjusted jointly within a dataset.

Patch-micro estimate. A metric computed over eligible patch predictions, with uncertainty clustered by partitioning unit.

Case-macro estimate. An equal-weight aggregation over partitioning units, limiting domination by heavily tiled cases. A slide is the aggregation unit only when it is itself the locked partitioning unit.

Self-contained search. The validation-search phase in which every candidate is fitted from scratch, needing no other method's result.

CE-inherited search. The validation-search phase for post-hoc logit adjustment and cRT, which reuse a condition's selected CE configuration and tune only what remains open for that method (τ for post-hoc logit adjustment; the stage-two learning rate for cRT), and so cannot begin until that condition's CE search has resolved.

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